

CHAPTER 15

HEREDITY: FROM PARENTS TO OFFSPRINGS

We have studied earlier that organisms reproduce to produce offsprings like themselves. The Babul(Acacia sp.) produces seeds that produce babul plants. The offspring of a goat resembles a goat while that of humans a human. Though offsprings resemble their parents they show some dissimilarity as well. Let us study the observable similarities and dissimilarities in humans and plants with the help of activities.

15.1 Similarities and dissimilarities in organisms

15.1.1 Variations in animals (Example: Humans)

Activity-1

Last year you had studied several similar and dissimilar characters among you and your friends. Now you shall study these within your family. Observe the characters given in the table among family members mentioned (those whom you have) and in you and put a mark ( ) wherever you find the character present in you and your family members.

Table-1

S.No.	Character		You	Mother	Father	Maternal grandpa	Maternal grandma	Paternal grandpa	Paternal grandma
1.	Colour of eye	Black Brown Green Any other (mention)							
2.	Dimple chin	Yes No							
3.	Rolling of tongue	Yes No							
4.	Finger next to toe	Longer than toe Shorter than toe							
5.	Ear lobe	Attached Free							
6.	Capable of bending the thumb backwards	Yes No							

Observe the filled table and answer the following questions-

- What are the characters present both in you and your parents?
- Are there any characters that you share with your parents and grandparents?
- What are the characters that are present in you and your grandparents but not in your parents?
- How do you think such characters may have passed on to you?

You may have observed that some characters have passed on to you from your parents and grandparents. Such characters that pass from one generation to the next are called as heritable characters and the study of the process in which they are transferred is called 'Heredity'. You have observed that some characters present in your family members were not present in you. The dissimilarities that we observe in spite of similarities are Variations about which we have studied in Chapter 1.

15.1.2 Variations in Plants

There are several similarities and dissimilarities in plants as well. Plants of the same species share several similarities like similar shape of flower or leaves but variations are such that even on the same plant it is impossible to find two same leaves (as you observed in your last class). Can you find two same seeds of a plant? Let's find out.

Activity-2

Observe the seeds in a pea or bean pod.

- Are there any seeds that are same?
- What are the variations that you see in them?

People have been observing variations in plants over a long period of time and developing plants of different races by selecting different characters. People had observed that characters of on generation pass on to the next or forthcoming generations.

But the question was how do different characters develop and how are they passed on from one generation to the next?

15.2 Heredity and Mendel's Contribution

Intensive studies started in the 19th century to find out how characters are passed on from one generation to the next. We studied in Chapter 1 that Darwin had observed different characters in organisms across generations and proposed the theory of origin of species by evolution. Though he observed characters passing from one generation to next he could not explain the mechanism of the process clearly. Darwin and many of his contemporary scientists were working towards understanding the mechanism among which the observations and contributions of Gregor Johann Mendel are most remarkable.

A monk in a monastery in Austria, Mendel, was inquisitive from childhood and interested in gardening. He would plant different plants in the garden of his monastery and try out different experiments on them. He conducted a series of around 10,000 well documented experiments for a period of nearly 12 years starting from the year 1856 . This lead to path breaking observations and a research paper which though not acknowledged in his time forms the basis of Heredity now. Even after that he carried on with his experiments to understand the process of heredity better and check if his previous observations were universally applicable. He documented all his observations and continuously communicated the same with his contemporaries. He would also read and analyzeddocuments written by other scientists of his time like Darwin who were performing similar experiments and writing about them. We came to know a lot about the process of heredity(that is transfer of characters from one generation to the next) with the help of Mendel's experiments. Mendel used mathematical calculations to hypothesize and confirm his results.

15.2.1 Mendel's experiments and their objectives

Mendelwas trying to look for a law that could be applicable to the hereditYof at least contrasting characters like yellow and green seed coat colour of pea seeds or white and purple colour of pea flowers as observed by him. He was also trying to look for a way to be able to predict how much variations may be observed from one generation to the next.



Gregor Johann Mendel

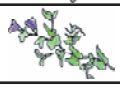
Mendel had started experimenting on bees but he quickly shifted to peas as plants were easy to control and observe. Moreover there were certain characters of plants of pea family that Mendel was already aware of.

He gave the following reasons for his choice of pea plants.

1. Peas are annual plants. Their life cycle being small, generations of pea plants may be observed and studied easily.
2. Pea flowers are usually self- pollinated where pollination is completed in the bud stage due to the presence of a protective crown over reproductive parts. This also helps to maintain characters of pea plants pure across several generations.
3. Cross pollination can thus be done artificially and acquire expected results as there is little chance of other pollen grains reaching the flower. Thus expected crosses or hybrids of pea plants can be obtained easily.
4. The plants are bisexual(having both male and female flower on the same plant, here in the same flower). If the male sexual part or stamens are removed, then the plant behaves as a unisexual one.
5. Hybrids formed are fertile and not sterile as found in most cases.
6. Peas have varieties that show clear contrasting characters.

Some contrasting characters that Mendel chose are given in the table-

Table - 2

S.No.	Type of Character	Clear contrasting varieties			
1.	Shape of seed	round		wrinkled	
2.	Color of seed	yellow		green	
3.	Color of flower	violet		white	
4.	Condition of pod	Swollen		Constricted	
5.	Colour of pod	Green		Yellow	
6.	Position of flower	Axial		Terminal	
7.	Length of plant	Tall		Dwarf	

He made certain arrangements for his experiments. He would allow self- pollination in some pea plants while cross pollination in others. He would continue experimenting with seeds obtained by cross pollination (that is, by the process of hybridization) in a separate manner as compared to those obtained by self -pollination. Each time that he obtained seeds in a particular experiment, he would count all of them and select and sow seeds of plants that showed a particular type of character. He would maintain records of all his experiments.

- Why did Mendel carry out cross-pollination to obtain hybrid seeds?
- Why was Mendel studying mostly contrasting characters?

15.2.2 Results of Mendels Experiments(with respect to crosses related to purple and white flower bearing plants)

Some pea plants bear white flowers while some others bear white flowers. Mendel selected such varieties of plants, obtained their seeds and grew more plants out of them. He now allowed self-pollination to take place in plants of a particular variety (that is - those with white flowers or those with purple flowers. Each time he selected flowers on the basis of their colour, he carried out this process for some generations till he could ensure that seeds of plants with purple flowers would yield only purple flowers while those with white flowers would obtain white. Thus by the process of repeated self-pollination, Mendel obtained plants that would produce seeds of purple or white varieties respectively for several generations. Mendel called these varieties pure. He started his experiments with these varieties of pure plants. He sowed seeds of pure varieties of white and purple plants. As the plants grew and started flowering he cross-pollinated them. The seeds obtained from cross pollinated flowers, or the hybrid seeds, yielded varieties that bore only purple flowers. This

generation of plants obtained from parental varieties of pure plants was first generation (F1 generation) plants.

- Think why intermediate varieties (those between purple and white) may not have been obtained?

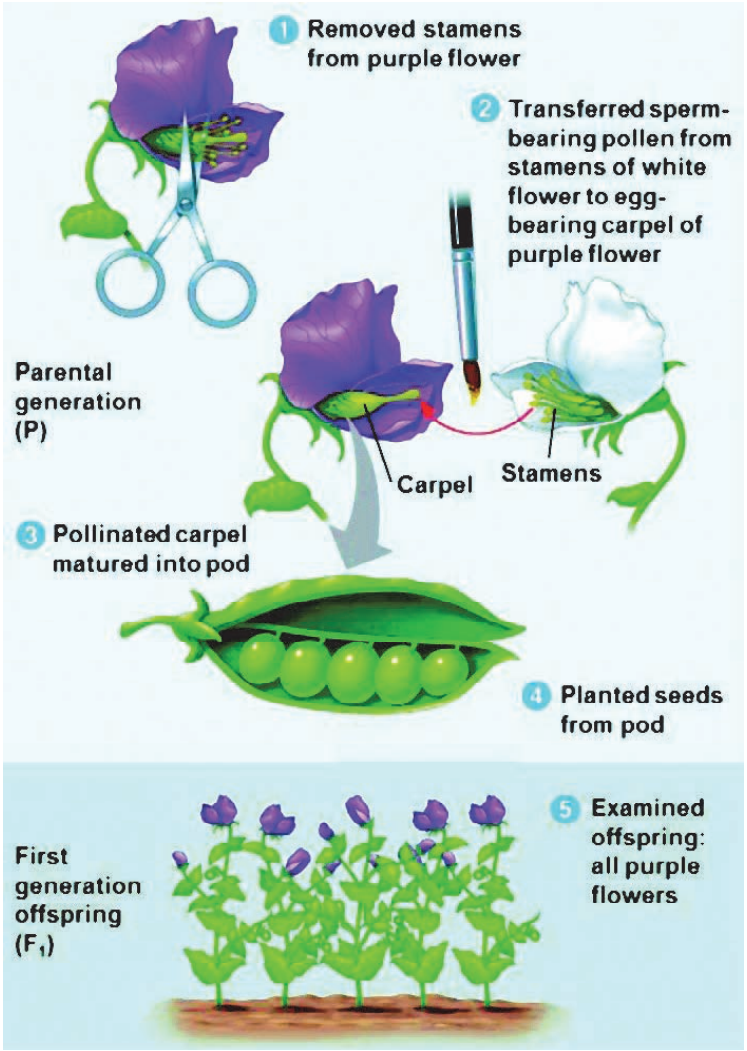


Figure-1: Cross pollination of pure parental types

Mendel now self- pollinated these first generation plants. The seeds obtained were then sown. To his utter surprise he found that these yielded plants of pure parental generation as well that he had started with. Not all plants yielded purple flowers. There were some that yielded white flowers as well. Among the 929 plants that had grown from the hybrid seeds, 705 of them bore purple flowers while the rest bore white flowers. Now if we see the ratio of purple to white varieties, it is 3.15:1.

Mendel studied several contrasting characters. He found that parental varieties with contrasting characters always gave a ratio of nearly 3:1 in the second generation.

Do you know?

The observations of experiments performed by Mendel-

Experiment	No. of Plants	Cross between Characters	F_1 Generation	F_2 Generation	Ratio
1	7324	Round and wrinkled	All round	5474 Round 1850 wrinkled	2.96: 1
2	8023	Yellow and Green seeds	All yellow	6022 yellow 2001 green	3.01:1
3	929	Purple and white flowers	All purple	705 violet 224 white	3.15:1
4	1181	Swollen Pod and Constricted pod	All swollen	882 swollen 299 constricted	2.95:1
5	580	Green and Yellow pods	All green	428 green 152 yellow	2.82:1
6	858	Axial and terminal flowers	All axial	651 axial 207 terminal	2.14:1
7	1064	Tall and dwarf	All tall	787 tall 277 dwarf	2.84:1

These observations lead Mendel to believe that the character, purple flower colour masked that of white completely in the first generation and partially in the second generation. Thus purple colour according to him was dominant and white recessive. He believed that expression of white colour in F2 generation could only happen if both characters were present independently in a hybrid seed.To find out how a character was expressed, Mendel took the help of mathematical calculation to figure out how this could be possible.

15.2.3 Assumptions based on experimental observations

Following are certain assumptions on the basis of experimental observations-

1. Each character is represented by two factors.
2. An offspring receives one factor of a character from parents. This is how an offspring always has a new pair of factors.
3. Among a pair of contrasting characters one is dominant and the other recessive.

Now, let us find out what was the mathematical basis of these assumptions.

15.2.4 Confirming assumptions with the help of Probability

Activity-3 The coin game

Let us take a pair of either one rupee, two rupee or five rupee coins. Mark both sides of one of the coins. See that the mark does not cover the head or tail portion of the coin completely (see figure for help).



Let this marked coin be 'A' and the one without marks be 'B'. Now toss both coins together and as they fall observe both and put a tally mark according to the condition you observe in the table given as under.

Table- 3

A	B	A	B	A	B	A	B
H	H	H	T	T	H	T	T

Repeat tossing the coin around 500 to a 1000 times. Now find out the percentage of all four conditions (HH,HT,TH,TT) as shown in table. Are you getting a value close to nearly 25% in all the said conditions? You would find that the ratio of all the four combinations of coins HH,HT,TH and TT is 1:1:1:1 (that is a probability of 25% each).

- If you would have played with the same coin what would be the probability of getting a head or a tail?

Thus if we assume that a coin represents a pair of Mendels' factors, and HH represents pure variety of purple coloured plants and TT represents pure variety of white coloured plants, then, HT, TH could represent a combination of characters, purple or white flowered plants. But Mendels' observation shows that nearly 75 percent(nearly 3 out of 4 parts showed by the ratio 3:1) bore purple flowers and around 25% plants bore white flowers.

To get a total of 75 we will have to add the percentages of HT and TH conditions to the HH condition. Thus starting with a pair of characters probability calculations started matching Mendel's observations. The presence of even a single purple factor leads to the expression of purple character. Mendel thus conducted several experiments with several contrasting characters to check his assumptions. Whether characters were represented by a pair of factors and among a pair of contrasting characters one was dominant?

Some more assumptions

As we have studied so far we know that Mendel prepared pure varieties of plants for a particular character and then started experimenting with them.

Mendel started his experiments with contrasting characters in the following manner (in the context of plants bearing purple and white flowers-

Assuming the characters of pure purple and white flower bearing plants to be represented by a pair of factors in each caseas under we get -

VV - purple

vv - white

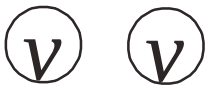
Their male and female gametes would be formed as-



Purple



White



Cross pollinating purple and white flowered plants we get-

	V	V
v	Vv	Vv
v	Vv	Vv

All plants of F1 generation have a combination of factors as-

These F1 generation plants would yield male and female gametes for F2 generation plants as-





The factors of second generation plants would be as-

	V	v
V	VV	vV
v	Vv	vv

This shows that the second generation yields 25% of plants each bearing purple and white flowers are like the parental varieties of first generation. While 50% of plants bearing purple flowers are like those obtained after F1 or the parental generation for F2.

Vv represents heterozygous factors while VV and vv represent homozygous factors.

Do you know?

We know now that factors that represent a character are called as genotype while the character that is expressed is called phenotype. But when the scientist A. Johannsen was proposing the use of these terms for the first time in the year 1909 in an article on the study of biodiversity, he wrote-'Gene is the unit of factors present in a cell formed by the fusion of male and female gamete while all genes together form the genotype.Organisms show a lot of similarities across generations. Phenotype is the variations caused due to effect of environment on organisms with same genotypic combinations. Thus genotypes are heritable characters while phenotypes are non-heritable characters.' We can see that the meaning and use of scientific terms change over time.

15.2.5 Mendel's Laws

Mendel observed that while the formation of gametes a pair of factors representing a character segregated independent of each other. Even when they formed a zygote by the fusion of male and female gametes they did not mix with each other. As for example the gametes of purple and white flower bearing plants would have a factor for purple and that of white respectively. The zygote formed by the fusion of male and female gametes would have a pair of factors for the type of colour. It could be VV or Vv for puple colour bearing plants or vv for white colour bearing plants. That factors segregate independent of each as gametes are formed and do not mix in zygote was Mendel's Law of segregation.

Mendel did not study just single pairs of contrasting characters but he combined two or more pairs of contrasting characters to see what would happen during the formation of gametes of such plants, whether the characters would mix or not. He also wanted to study whether characters influenced each other. The discussion of Mendel's cross with more than a single pair of character is given in annexure to this chapter.

Do you know?

Mendel's observations of crosses of similar and contrasting characters formed the basis of heredity. Nobody during his time could understand the importance of his work. It was about 30 after his death that scientists working in the area of heredity could recognize and appreciate the immense importance of his work.

We do know now that mixing of characters do take place and that Mendel's laws may not be applicable completely as such.For example in organisms that reproduce by asexual reproduction and several of those where they do not have factors in pairs (Mustard where chromosomes are quadrupeds and wheat upto dodecapeds or codominant characters as in human blood group).

15.3 Human inheritance on the basis of Mendel's laws

We had studied a few characters in activity 1. A Mendellian character or that which exactly follows Mendel's law is 'dimple chin'. Sickle cell anaemia, a dreaded condition of heritable characters is an example of a character that follows Mendellian inheritance.

15.3.1 Factors for Sickle cell anaemia and their inheritance

The 11th pair of autosomal chromosome(see page….) has factors responsible for formation of the protein haemoglobin.

- What is the function of this protein(refer to chapter 7 for the answer)?

The presence of these factors help in the formation of normal haemoglobin and normal disc like red blood cells. To show a normal pair of factors we have used the symbol RR(representing normal haemoglobin) in figure 3. A change in the composition of these factors leads to a change in

composition of haemoglobin (abnormal haemoglobin)leading to the formation of sickle shaped red blood corpuscles. A condition represented by 'rr' Sickle celled blood cells have very little oxygen carrying capacity. Moreover the life cycle of such cells are 15 to 20 days only while that of a normal blood cell is 100 to 120 days. Patients suffering from sickle cell anaemia become tired and fatigued quickly. It becomes fatal when the factors are homozygous for the recessive character 'rr' as shown in the figure. The factor forming normal haemoglobin is dominant over that of abnormal type.

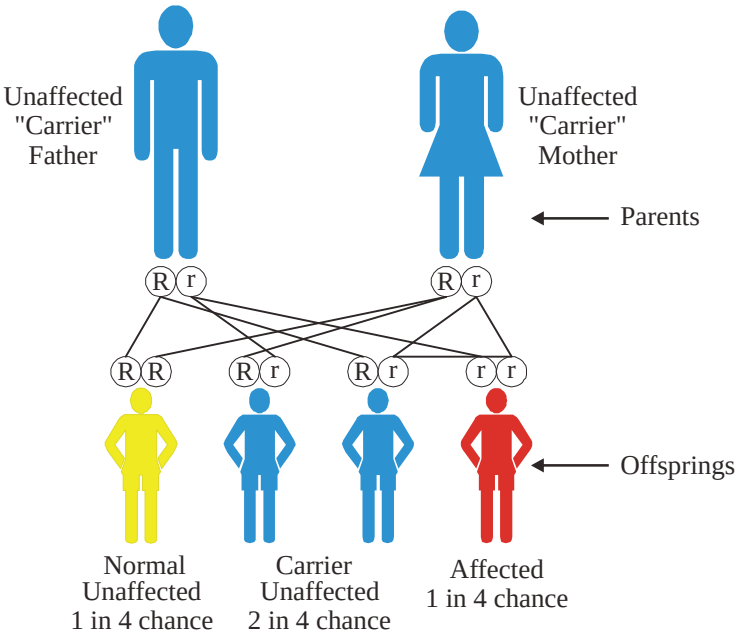


Figure-2

Do you know?

Sickle cell anemia may also be a favourable character. Those that are carriers of the character are have developed resistance towards the malarial parasite as it fails to colonize in the sickle shaped blood cells. The incidence of sickle cell anemia is seen in places that also have a high incidence of malaria.This is not only found in our state but in several other parts of our country as well as the world especially in certain African countries.

15.3.2 Sex determination in Humans

We have studied about human chromosomes in the previous chapter. We also studied that a pair of sex chromosomes (the 23rd pair of chromosomes) are responsible for sex determination. A female carries a homozygous pair (XX) while males have a heterozygous one (XY, Y being shorter than X).

Gametes as we know carry only one set (haploid) of a pair of set of chromosomes and after the fusion of male and female gamete, the zygote so formed contains a diploid pair of chromosomes. The sex of the child formed from this zygote depends on whether it has a pair of XX chromosomes or has XY chromosomes in the 23rd pair.

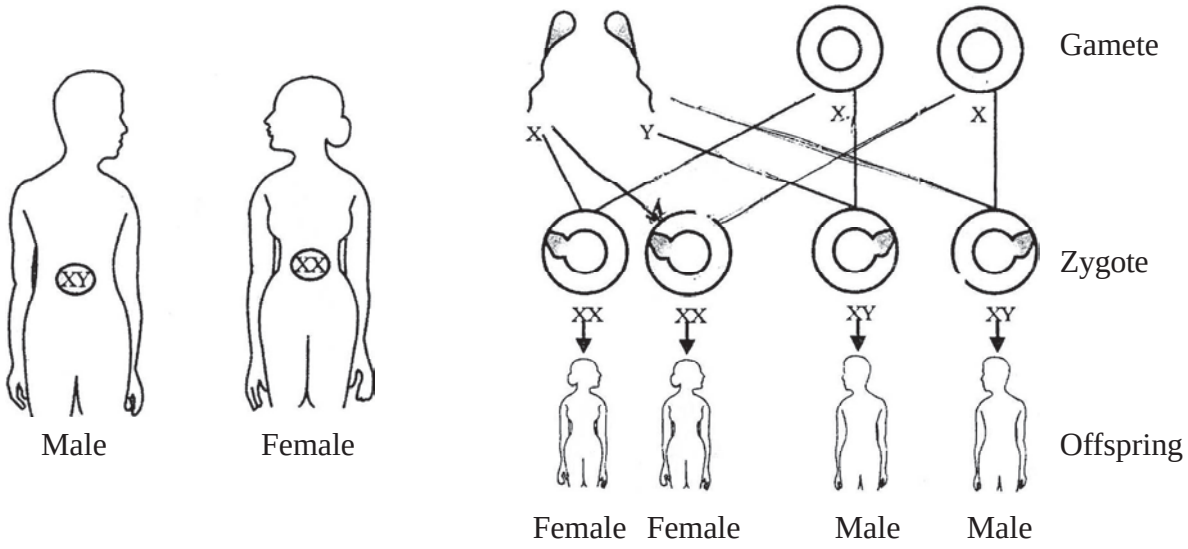


Figure-3: Sex determination in Humans

- See Figure 4 what do you think are the probabilities of each of the four conditions XY,XY, XX and XX?
- What do you think is the probability of having a female child?

We can understand from the figure that the probability of having a male or a female child is 50% each.

Thus having a male or a female child is just one out of two chances. None of the parents are responsible for the same.

- "The human male carries the Y chromosome so to have a male child a human male might have some role." Do you agree with this statement? Why/why not?

15.4 Factors to genes:

Mendel's contribution to heredity forms the basis of our knowledge of the same. He is thus called father of genetics. Mendel had hypothesized that characters are expressed by a pair of factors. Nearly a 60 years after Mendel's findings, in the year 1920, Sutton, a scientist working on the chromosomes of locust showed that Mendellian factors were located on chromosomes and only one chromosome of a pair of parental chromosome passes to the offspring.

If we consider Mendel's purple and white coloured flowers of pea plants, we find that there was no intermixing of characters. It was found that they are located on different chromosomes of a pair representing the colour and there is no shuffling of characters.

Mendel's factors later came to be known as genes. Genes were considered as the units of heredity. We also came to know later that expression of purple flower colour of peas is due to a single gene.

But what are genes? As research in the area extended further and the genetic material DNA(deoxy ribo nucleic acid) was discovered, we came to know that gene was that part of DNA that carried information for the formation of protein. When even a single gene is shuffled the whole group of genes changes and a new combination is formed. Offspring with such a group of genes contain a new genetic combination.

Concept of species on the basis of formation of new genetic combinations is often referred to as genetic species concept.

The whole human genome has been mapped andit is now being used for many purposes like the identification and cure of genetic diseases.

Key Words

Heredity, contrasting characters, dominant, recessive, homozygous, heterozygous, carrier.



What we have learnt

- The transfer of characters across generations is called heredity.
- Inspite of similarities the differences in characters that are present in members of a species of organism is called as variation.
- Those variations that pass from one generation to the next like -dimple chin, hair colour are heritable variations.
- Those characters which do not pass from one generation to the next like mark of a wound, obesity etc. are called as acquired characters.
- Mendel performed experiments for several years to understand how characters were passed from one generation to the next. He extensively experimented mainly with pea plants provided us with some fundamental laws of heredity.
- He used mathematical deductions(mainly probability and chance) to test his hypothesis and deduce his laws.
- We come to know about the following about heredity from Mendel's experiments-
- Each character is expressed by a pair of factors
- During reproduction offspring receives only one of a pair of factors from a parent.
- Contrasting pairs of characters are represented by pairs of contrasting factors
- Law of Segregation: Factors do not mix but are passed on to gametes independently.
- Sickle cell anaemia follows Mendellian inheritance. The composition of haemoglobin changes, thereby changing the shape of normal red blood cells from disc shaped to sickle shaped.
- Zygotes containing XX as the 23rd pair of chromosome form females while those containing XY form males. The probability of having a female or male child is 50% for each case.

Assessment

1. Choose the right option-

(i) What do you think may have been the reason for Mendel's success?

(a) Clear planning based on reasoning, experimenting, recording observations etc. and duly writing article regarding the same.

(b) Focusing on certain clear contrasting characters.

(c) Use of mathematical calculations.

(d) All above options

(ii) Mendel had studied certain contrasting characters of pea plants. Which character do you think was not chosen by Mendel?

(a) Tall and dwarf plants (b) Yellow and green coloured seeds

(c) Terminal and axial flower (d) Round and wrinkled stem

(iii) Some people can roll their tongue (roller represented by RR/Rr) and this is an autosomal dominant character. Those who can't roll have the recessive factors(Non roller represented by rr)

A child is capable of rolling her tongue. Her brother fails to roll the tongue while two of her sisters can roll their tongues. If both of the parents can roll their tongue, what do you think would be the combination of factors in them?

(a) RR and RR (b) Rr and Rr (c) RR and rr (d) rr and rr

2. Let us consider 4 major blood groups of humans A, B, AB and O. These had been named on the basis of certain antigenic factors.. The presence of both A and B factors represented AB blood type while absence of both represented O blood type. Let us take I to represent the substance responsible for the said factor. IO then represents absence of the substance. Now observe the following table and answer the questions given below the table.

Blood Group	Factors
A	I^AI^A, I^AI^O
B	I^BI^B, I^BI^O
AB	I^AI^B
O	I^OI^O

- Those having blood group A would have how many IA factors?
- Those of blood group O should have how many IO?
- Which factor do you think is recessive . IO, IA or IB?
- 'A recessive character represents pure parental generation.' Justify this statement.
- A child of parents having blood group A has O group. What do you think are the factors that the parents have? What do you think could be the probable blood group type of a second child that they have?

3. When a blue flowered plant (Bb) is crossed with a white flowered plant (bb) what are the combinations possible in offspring generation?



4. A variety of cow named 'Zeba' is adapted to survive in hot conditions. Another variety 'Sahiwal' gives 20,000 liters of milk every year. The variety 'Angush' has a well built body. If we want to acquire maximum milk from a variety of Cow here in Chhattisgarh, which varieties of cows do you think must be crossed? Give reasons for your answer.

5. How will our knowledge of heredity help in improving our crop plants to acquire maximum production?

6. How can we choose suitable characters to obtain new varieties of plants? (Hint: We had studied about plants of mustard family in Chapter 1, you may site examples of other plants)

7. A farmer sowed seeds of purple flowered pea plants and claimed that all plants in the following generation would have purple flowers. Was he right? Why, why not?

8. We have been able to successfully obtain pink flowered Rose plants by crossing white and red coloured Rose plants respectively. .Do you think that some characters do intermix? Why? Site some other examples of this type that are related to agriculture.

9. What would happen if you played the game of coins with just one coin? What is the probability of getting a head then?

10. Can we develop high yielding variety seeds by hybridization? Can such seeds be developed only by crossing pure varieties?

11. The sweetness of milk is due to the presence of Lactose (a type of sugar) in it. You may have heard about people who fail to digest milk and milk products. The factor responsible for the production of the enzyme capable of digesting lactose is usually not present in these people. Even if a single factor for formation of the enzyme is present, lactose digesting enzyme is formed. Now answer the following questions on the basis of this information-

    - What would be the combination of factors of a child who fails to digest milk and milk products? What are characters represented by such factors called?
    - What is the combination of factors in the parents of such a child(take a lactose digesting factor as L)who are capable of digesting milk?
    - What percent of children of such parents would be able to digest milk?

12. In green gram, the development of two flowers at the axial position is a recessive character, while the development of a single flower at the axial position is a dominant character. If the dominant condition is represented by 'SS' and recessive by 'ss' then answer the following questions -

    - Find out the percentages of pure and hybrid plants obtained in the second generation(F2 generation) after crossing pure parental varieties.

- Find out the percentage of each variety of offsprings obtained by a cross between Heterozygous Ss and pure ss varieties.
- What would be the percentage of varieties with single axial flowers when a cross between homozygous variety SS and heterozygous variety Ss is conducted?

Show the formation of gametes and combination of factors in each of the questions of this section.

Annexure

Dihybrid cross and another Law of Mendel-

Mendel had not only observed the inheritance of single characters, he had also observed characters in combinations. As for example Mendel observed crosses of plants that were grown from such characters as round and yellow seeds(RRYY) and wrinkled and green seeds(rryy).

He started with pure parental combinations(of nearly 600 seeds) and found that all plants obtained at the end of F1 generation bore round yellow seeds. When plants were grown from such seeds, he observed combinations like 315 round yellow, 108round green, 101 wrinkled yellow and 32 wrinkled green. The ratio of 9:3:3:1 obtained could only be possible if the hybrid varieties RrYy formed gametic combinations in the form - RY, Ry, rY and ry respectively.

Law of Independent assortment - Mendel concluded that different characters assort independently of each other while formation of gametes and form new combinations in forthcoming generations. Mendel deduced this law by observing inheritance of more than one character at a time. The following diagram shows a schematic representation of results obtained after a dihybrid cross(crosses obtained by observing two characters at a time in parental varieties)

